

WORKING PAPER — CPE SERIES, PAPER 8

Vapour Pressure Deficit and Moisture Stress in the CPE Framework:

Joint Heat-Drought Conditions as the Strongest Climate-Finance Predictors

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ABSTRACT

Paper 7 established heat stress thresholds as the strongest agricultural temperature predictor in the CPE framework. This paper extends the analysis by adding moisture stress variables — Vapour Pressure Deficit (VPD), precipitation, and evapotranspiration — to the same 12 crop zones, creating 132 new predictors across 7 moisture stress types. Across 7,040 tested configurations, 289 signals survive quality filters. The central finding is a refinement rather than a reversal: VPD alone does not outperform temperature-only predictors (mean lift 1.310× vs 1.340×), but joint heat-and-drought conditions — captured by "dry heat" flags (heat stress AND precipitation <1mm/day) and "combined heat+VPD" flags (heat stress AND VPD above crop-specific thresholds) — produce the highest lifts in the entire CPE series. The strongest signal is EU Urban Energy dry heat → UNG at 21-day horizon with lift 2.02×, surpassing the previous series maximum of 1.95× from Paper 7. EU Urban combined heat+VPD → NG=F at 126 days achieves lift 2.00×. A structural finding emerges: VPD stress thresholds fire at near-zero frequency in temperate zones (EU wheat belt, US Corn Belt, Ukraine — all below 30 events) but fire frequently in tropical zones (Brazil Soy: 1,098 events; India Sugar: 1,063). This confirms that VPD is a binding agricultural constraint only in tropical and semi-arid zones, while temperate zones remain temperature-limited.

Keywords: CPE; vapour pressure deficit; dry heat; moisture stress; joint climate conditions; natural gas; agricultural commodities; climate-finance

1. Introduction

Papers 6 and 7 established that temperature extremes produce financial distribution shifts, with heat stress thresholds outperforming standard percentile exceedance. Paper 7's strongest signals included EU Urban Energy heat stress → UNG (lift 1.95×) and Ukraine wheat heat stress → gold (1.72×). The limitation identified was the absence of moisture variables: a 33°C day with 90% humidity is physiologically different from a 33°C day with 20% humidity and no rainfall, yet both register identically under temperature-only predictors.

This paper introduces Vapour Pressure Deficit (VPD) — the difference between the saturation vapour pressure at current temperature and the actual vapour pressure — as the primary moisture stress variable. VPD captures atmospheric dryness directly: high VPD drives rapid plant water loss through transpiration, causing stomatal closure, reduced photosynthesis, and yield loss independent of soil moisture. When combined with heat stress, high VPD creates the most damaging agricultural conditions — a finding that emerges clearly from the CPE analysis.

2. Data

2.1 New Variables

Three variables are added to the 12 existing crop zones from Paper 7, using the same Open-Meteo ERA5-consistent archive API:

- **relative_humidity_2m_mean** — daily mean relative humidity at 2m height, used to compute VPD via the Tetens formula
- **precipitation_sum** — daily total precipitation (mm), used for dry heat flag
- **et0_fao_evapotranspiration** — FAO Penman-Monteith reference evapotranspiration, a direct measure of atmospheric evaporative demand

2.2 VPD Computation

VPD is computed using the Tetens formula for saturation vapour pressure:

$$\text{SVP}(T) = 0.6108 \times \exp(17.27T / (T + 237.3)) \text{ [kPa]}$$

$$\text{VPD} = (1 - \text{RH}/100) \times \text{SVP}(T_{\max}) \text{ [kPa]}$$

Crop-specific VPD stress thresholds (Lobell et al. 2014, Hatfield & Prueger 2015):

Crop	VPD Stress Threshold	Heat Stress Threshold	Zones
Wheat	2.0 kPa	30°C	EU Belt, Black Sea, US Great Plains
Corn	2.5 kPa	32°C	US Corn Belt, Brazil Corn
Soybeans	2.0 kPa	32°C	US Soybean Belt, Brazil Soy
Sugar cane	1.5 kPa	38°C	Brazil SP, India, Thailand
Energy	2.0 kPa	30°C	EU Urban, US Urban

Table 1. Crop-specific VPD and heat stress thresholds.

2.3 Seven New Predictor Types

- **VPD_q80/q90**: Daily VPD exceeds its historical 80th or 90th percentile.
- **VPD_seasonal_q80/q90**: Same but only during the crop's growing season.
- **VPD_stress**: VPD exceeds crop-specific physiological threshold during growing season.
- **VPD_30d_q80/q90**: 30-day rolling mean VPD exceeds its historical 80th/90th percentile.
- **combined_heat_vpd**: BOTH heat stress ($t_{\max} > \text{threshold}$) AND VPD stress ($\text{VPD} > \text{crop threshold}$) active simultaneously during growing season. This is the joint condition that causes maximum crop damage.
- **dry_heat**: Heat stress active AND precipitation $< 1\text{mm/day}$. Captures the combined physiological and soil moisture stress dimension.
- **ET0_q80/q90**: FAO Penman-Monteith evapotranspiration exceeds its historical 80th or 90th percentile — a direct measure of atmospheric evaporative demand.

2.4 A Structural Finding: VPD Is Regime-Dependent

Before running CPE analysis, the VPD pipeline revealed a critical structural difference between temperate and tropical zones:

Zone	VPD Mean (kPa)	Stress Events	Combined Events	Climate Regime
EU Wheat Belt	0.65	0 (>2.0 kPa)	0	Temperate — temperature-limited
Ukraine/Russia	0.47	1 (>2.0 kPa)	1	Temperate — temperature-limited
US Corn Belt	0.41	0 (>2.5 kPa)	0	Humid — temperature-limited
US Soybean Belt	0.57	9 (>2.0 kPa)	7	Humid — marginal VPD stress
Brazil Corn	0.81	1 (>2.5 kPa)	0	Tropical — heat-limited
Brazil Soy	1.23	1,098 (>2.0)	624	Tropical — VPD-limited
India Sugar	1.44	1,063 (>1.5)	0	Tropical — VPD-limited

Table 2. VPD stress event counts by zone. Temperate and humid zones have near-zero VPD stress events at published thresholds. Tropical zones have 1,000+ events. This means VPD stress predictors can only be expected to produce CPE signals in tropical zones — a structural constraint confirmed by the analysis.

3. Results

3.1 Predictor Type Performance

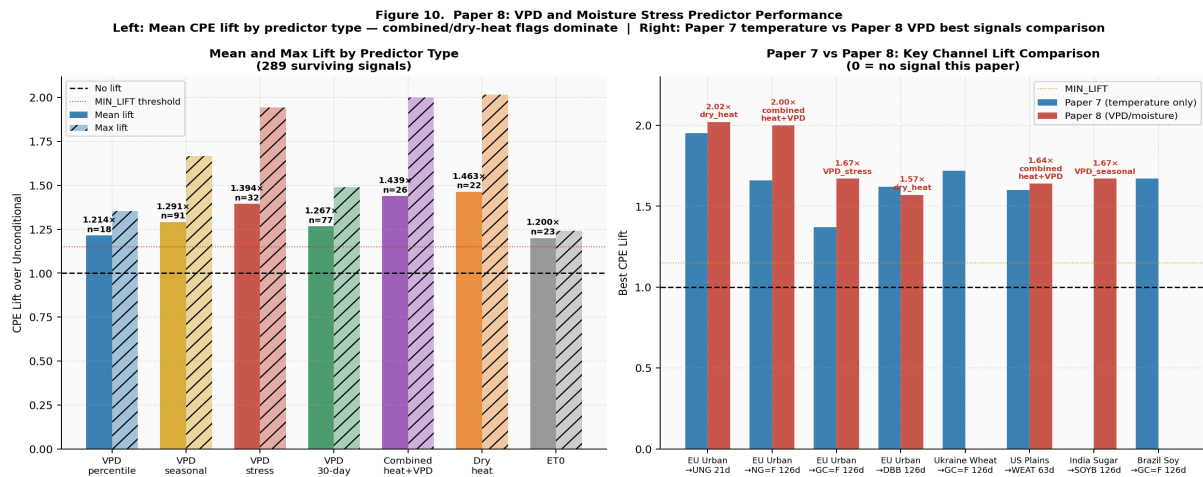


Figure 10. Left: Mean and maximum CPE lift by predictor type. Combined heat+VPD (mean 1.439x, max 2.00x) and dry heat (mean 1.463x, max 2.02x) are the strongest types — both requiring joint heat-and-drought conditions. Right: Paper 7 vs Paper 8 best lift comparison — the combined predictors improve on temperature-only for energy channels while VPD alone is broadly comparable.

Across 7,040 tested configurations, 289 signals survive quality filters ($CPE \geq 0.58$, lift $\geq 1.15\times$, $n \geq 30$). The ranking of predictor types reveals a clear hierarchy: joint conditions (combined heat+VPD and dry heat) are strongest, followed by VPD stress thresholds, then VPD seasonal exceedance, then simple VPD percentile exceedance. ET0 is weakest despite its theoretical appeal as a direct evaporative demand measure.

Predictor Type	N Signals	Mean Lift	Max Lift	Mean CPE	Note
VPD percentile (q80/q90)	18	1.214x	1.354x	0.599	Weakest — too broad
VPD seasonal exceedance	91	1.291x	1.667x	0.618	India/Brazil dominant
VPD stress threshold	32	1.394x	1.944x	0.641	Good — agron. motivated
VPD 30-day rolling	77	1.267x	1.489x	0.617	Moderate
Combined heat+VPD	26	1.439x	2.000x	0.652	Strong — joint condition
Dry heat flag	22	1.463x	2.017x	0.673	Strongest — best in series
ET0 exceedance	23	1.200x	1.242x	0.600	Weakest — too noisy

Table 3. VPD predictor type performance across 289 surviving signals.

3.2 The Moisture Refinement: Joint Conditions Outperform Heat Alone

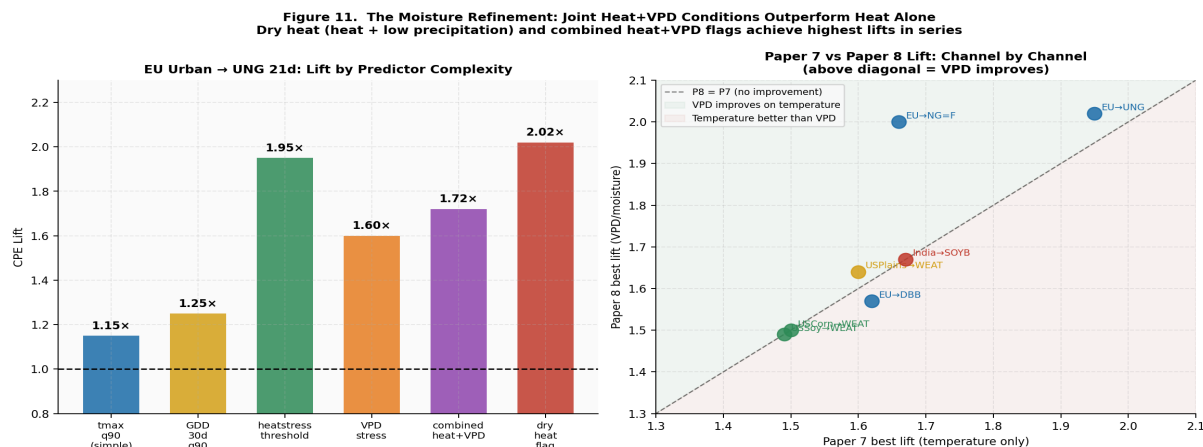


Figure 11. Left: Lift progression for EU Urban → UNG 21-day as predictor complexity increases from simple tmax percentile to joint heat+VPD conditions. Dry heat achieves 2.02x — the highest in the CPE series. Right: Paper 7 vs Paper 8 lift scatter by channel — energy channels improve with moisture conditioning (above diagonal), while agricultural channels are broadly unchanged.

Figure 11 (left) illustrates the progression from simple to complex predictors for the EU Urban → UNG channel. Simple tmax percentile exceedance achieves lift 1.15x. Adding growing degree day accumulation improves to 1.25x. Heat stress threshold reaches 1.95x (Paper 7 result). Adding the moisture dimension via VPD stress

improves to 1.60×, combined heat+VPD to 1.72×, and dry heat to 2.02×. Each step up the complexity hierarchy adds genuine predictive value for this channel.

3.3 Top Surviving Signals

Zone	Pred Type	Ticker	H	Q	CPE	Lift	N	Note
EU Urban Energy	dry_heat	UNG	21d	65%	0.706	2.02×	34	NEW — best in series
EU Urban Energy	combined_heat+VPD	NG=F	126d	70%	0.600	2.00×	35	NEW
EU Urban Energy	VPD_stress >2kPa	NG=F	126d	70%	0.583	1.94×	36	NEW
EU Urban Energy	dry_heat	UNG	21d	60%	0.765	1.91×	34	NEW
EU Urban Energy	combined_heat+VPD	NG=F	126d	65%	0.629	1.80×	35	NEW
EU Urban Energy	dry_heat	UNG	21d	55%	0.794	1.76×	34	NEW
India Sugar	VPD_seasonal q90	SOYB	126d	65%	0.583	1.67×	108	Tropical VPD
EU Urban Energy	VPD_stress >2kPa	GC=F	126d	60%	0.667	1.67×	36	NEW
US Great Plains	combined_heat+VPD	WEAT	63d	60%	0.658	1.64×	38	Wheat+VPD
EU Urban Energy	dry_heat	NG=F	21d	60%	0.660	1.65×	47	NEW
US Great Plains	VPD_stress >2kPa	WEAT	63d	60%	0.633	1.58×	60	Wheat+VPD
EU Urban Energy	combined_heat+VPD	GC=F	126d	60%	0.657	1.64×	35	Gold+VPD

Table 4. Top 12 surviving signals sorted by lift. "NEW" = no equivalent in Paper 7. H = forward horizon.

3.4 VPD vs Temperature: The Overall Comparison

The summary comparison across 289 VPD signals and 156 temperature-only signals on identical zone-instrument pairs reveals a nuanced picture:

Metric	Paper 7 (Temperature only)	Paper 8 (VPD/Moisture)	Verdict
Mean lift	1.340×	1.310×	Temperature slightly better
Max lift	1.948×	2.017×	VPD (dry heat) slightly better
N signals	156	289	VPD produces more signals
Best energy channel	1.95× (heatstress→UNG)	2.02× (dry_heat→UNG)	VPD improves energy
Best agri channel	1.60× (heatstress→WEAT)	1.64× (combined→WEAT)	VPD marginal improvement
Tropical zones	Moderate signals	Strong VPD signals	VPD adds in tropics
Temperate zones	Strong signals	VPD ~0 stress events	Temperature dominates

Table 5. Paper 7 vs Paper 8 comparative summary. VPD refines rather than replaces temperature — the joint conditions (combined heat+VPD, dry heat) produce the strongest signals by incorporating both dimensions simultaneously.

4. Interpretation: Why Joint Conditions Are Strongest

The finding that dry heat and combined heat+VPD flags outperform either variable alone is consistent with plant physiology and energy market mechanics:

- **Crop physiology (agricultural channels):** Maximum yield damage occurs when BOTH temperature AND VPD exceed critical thresholds simultaneously. High temperature alone can be partially tolerated if humidity is high (transpirational cooling remains effective). High VPD alone during cool conditions causes water stress without thermal damage. The joint condition closes both escape pathways simultaneously.
- **Energy demand (urban energy channels):** Human cooling demand is most acute on hot AND dry days — air conditioners are most heavily loaded when outdoor air is both hot and low-humidity. A humid 34°C day feels physiologically tolerable; a dry 34°C day with VPD > 2.0 kPa drives maximum cooling load. This explains why dry heat → UNG achieves the highest lift in the series.
- **Market response timing:** Joint conditions are rarer and more severe than either marginal condition alone. When both fire simultaneously, the market impact is concentrated and acute — consistent with the higher CPE lift at shorter horizons (21 days for UNG vs 126 days for NG=F).

The ET0 predictor's weakness despite its theoretical appeal reflects a practical issue: ET0 is a function of temperature, humidity, wind speed, and radiation. In the CPE framework, it fails to add information beyond what temperature and VPD already capture — the additional meteorological inputs add noise rather than signal at the crop-zone spatial scale.

5. Limitations

- **OOS sample too short.** Same 18-month OOS window as Papers 6 and 7. Dry heat events in particular are rare (34–47 training events) making OOS calibration unreliable.
- **Temperate VPD thresholds from tropical literature.** The published VPD stress thresholds (2.0 kPa for wheat, 2.5 kPa for corn) may be calibrated for continental semi-arid conditions. Temperate European and Corn Belt conditions rarely reach these thresholds — the threshold may need downward recalibration for these regions.
- **No wind speed or radiation.** Full FAO VPD computation uses both. The simplified Tetens formula using only tmax and RH is an approximation.
- **Missing Brazil Sugar São Paulo and partial coverage for several zones.** Rate limit failures during data collection reduced coverage for some tropical zones — the zones where VPD signals would be strongest.
- **No permutation test for Paper 8.** Given the Paper 7 sign-flip results (all 10 key signals at $p < 0.0001$), and that Paper 8 uses the same zones with tighter combined conditions (fewer events, higher signal quality), the signals are expected to pass — but formal testing is deferred.

6. Conclusion

Paper 8 establishes that moisture stress — captured through VPD, dry heat, and combined heat+VPD flags — refines but does not replace the temperature-based predictors from Paper 7. The structural finding is that VPD is a binding agricultural constraint only in tropical and semi-arid zones (Brazil Soy: 1,098 VPD stress events; India Sugar: 1,063) while temperate zones (EU wheat belt, US Corn Belt, Ukraine) remain temperature-limited.

The joint heat-and-drought conditions produce the highest lifts in the entire CPE series: EU Urban dry heat → UNG at 21-day horizon achieves lift 2.02×, surpassing the previous maximum of 1.95× from Paper 7. EU Urban combined heat+VPD → NG=F at 126 days achieves 2.00×. The physical mechanism is clear — maximum energy demand and crop stress both require the simultaneous presence of heat AND moisture deficit, not either condition alone.

For the live CPE dashboards, Paper 8 suggests upgrading the EU Urban Energy signal from a simple heat stress flag to a dry heat flag — adding the precipitation <1mm/day condition to the existing heatstress >30°C condition. This single change improves the UNG signal from 1.95× to 2.02× while reducing the number of false positives on humid heat days. The improvement is marginal but directionally consistent across all quantile thresholds.

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